The Impact of Placement Exams on Refugee Children's School Enrollment and Grade Assignment

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Abstract

While placement exams can help assign refugee children to the right grade in host country schools, poor implementation may turn them into a barrier to school access. This paper studies the effects of mandatory placement exams prior to school enrollment for Venezuelan refugee children in Northern Brazil. Using national administrative enrollment data, we estimate difference-in-differences and triple difference models exploiting the policy's geographic rollout and its targeted age range (8–14). We find that the policy reduced refugee enrollment by 15% between 2020 and 2024. Asylum application data suggest that affected refugee children are likely out of school, with no offsetting enrollment gains observed elsewhere in the country. Exam registry data reveal that the main barrier was not registration failure but high absenteeism (30%), driven by long waiting times, difficulties notifying families of exam dates, and insufficient exam availability beyond the conventional enrollment month. Placement exams also increased grade downgrading among enrolled refugees, exacerbating age-grade distortion.

Keywords: Refugee Children; Access to Education; School Enrollment; Placement Exams; Venezuelan Migration

JEL Codes: I25, I28, J15, O15, F22

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1 Introduction

Between 2010 and 2024, the number of displaced children worldwide almost tripled, rising from 17 million to almost 50 million - UNICEF. This group includes migrants, displaced persons, asylum seekers, and refugees. For these children, access to education in the host country not only provides a pathway to future opportunities and integration, but also offers stability, psychosocial support, and protection - Bridges and Walls (2018).

However, host country schools can face multiple challenges when enrolling displaced children and assessing their skills to determine appropriate integration strategies, such as grade placement.[1] First, language differences often pose an immediate barrier. Second, displaced children often experienced prolonged interruptions in schooling. Third, displaced families may flee without or lose academic transcripts and records. Finally, enrollment and attendance may be adversely affected by high outside options arising from families' socioeconomic vulnerability and the prevalence of child labor.

These challenges highlight the importance of creating a well-developed reception system for immigrant students that promotes integration, enrollment, and sustained attendance.[2] In this context, placement exams can help assess students' pre-enrollment knowledge and guide integration strategies into schools. However, depending on their implementation design, they may also function as an unintended extra burden to enrollment.

This paper examines how the introduction of a mandatory placement exam affects refugee children's school enrollment. We study the sudden and geographically concentrated inflow of displaced Venezuelans into Brazil triggered by Venezuela's economic and political crisis. In 2018 alone, more than 150,000 Venezuelans entered Brazil through the border state of Roraima. In Boa Vista, Roraima's capital and the main destination for migrants after crossing the border, the enrollment of Venezuelans jumped from virtually 0% in 2016 to almost 20% of the total student population in 2024. At the end of 2019, to standardize the enrollment process of foreign students,

[1]Kirdar, Koç, and Dayioglu-Tayfur (2021) finds that the enrollment gap between Syrian refugees and natives in Turkey persists even after controlling for a range of socioeconomic characteristics and is particularly pronounced among children who arrived after age 8.

[2]See policy reports: Luciak (2012) (European Union) and OECD. (2019) (OECD).

the local government implemented a mandatory pre-enrollment placement exam for immigrant children aged 8 to 14. To schedule the exam, families were required to call a designated phone number or visit a government office, after which they would receive a return call confirming their exam date — a process that, on average, took 30 days.

Using the national administrative enrollment data from 2017 to 2024, we employ a triple-difference framework that exploits both the policy's geographic implementation (exclusively in Boa Vista) and its targeted age range (exclusively 8 to 14). We find that the policy reduced school enrollment of affected refugee children by 30% between 2020 and 2024. We find no differential effects by gender; however, the decrease in enrollment was concentrated among ages 8 to 11, since 12 to 14-year-olds could enroll in state-managed elementary schools that did not require the exams. The event study shows no differential pre-trends, and a placebo simulation across the 53 municipalities most exposed to Venezuelan students places Boa Vista's estimate in the extreme left tail of the distribution. We further verify using asylum application data from CONARE that the age composition of the Venezuelan refugee population in Boa Vista did not shift differentially in the targeted age group after 2019, ruling out the most natural demographic alternative explanation.

The administrative data on the placement exam registry allows us to open the black box of why enrollment fell. The placement exam process involved two steps: registering on the wait list and attending the exam. Using registry data from 2023 and 2024, we show that registration itself was not the binding constraint: the number of registered children would have been more than sufficient to close the estimated enrollment gap. The main barrier was exam absenteeism (30%). More specifically, the waiting time (on average, 30 days) and difficulty contacting families (some with only NGO phone numbers in their registry) increased the chances of exam absenteeism. Finally, Venezuelan children living in shelters, a specially vulnerable subgroup of the refugee population, were 20 percentage points more likely to miss/don't attend the exam.

The results suggest that implementation capacity is critical to avoiding adverse effects on refugees' access to education. Therefore, prioritizing enrollment over the placement exam can help mitigate the policy's negative effects. Importantly, improving exam capacity (i.e., reducing waiting times) and communication (e.g., publicly available exam schedule lists and direct administration of exams in refugee shelters) can reduce absenteeism and increase enrollment. These recommendations are directly relevant to other host-country governments managing large-scale refugee inflows, and align with UNHCR and OECD guidance emphasizing that reception system design, not just policy intent, determines whether displaced children access education in practice.

This paper contributes to the limited literature on immigrants' integration into host-country schools.[3] Conger (2013) finds that English language learners in Miami-Dade County who were assigned to lower grades experienced improved outcomes in mathematics but showed no gains in reading or retention. Höckel and Schilling (2022) finds that immigrant children's enrollment in preparatory language-learning classes led to lower test scores than direct placement in regular classrooms in Germany. Another set of papers studies how school tracking (vocational and academic tracks) affects second-generation immigrants' performance and attainment in Europe - see Lüdemann and Schwerdt (2013), Jang and Brutt-Griffler (2023), and Cobb-Clark, Sinning, and Stillman (2012). This study examines a different policy and its impact on enrollment, a particularly relevant outcome in developing settings.

In addition, our mechanism results connect to other papers showing that removing barriers to exam access can substantially increase take-up, even when exams are high-stakes - Bulman (2015) (opening nearby SAT centers), Pallais (2015) (giving ACT takers more free score reports), and Muralidharan and Prakash (2017) (facilitating transportation to schools in India).[4] Our contribution is to show that the same logic operates in reverse: when exam access costs are imposed rather than reduced, enrollment falls sharply, and the costs fall most heavily on the most vulnerable.

The rest of the paper proceeds as follows. Section 2 describes the Venezuelan refugee inflow into Brazil, the structure of Brazil's public education system, and the policy change in detail. Section 3 describes the data. Section 4 presents the empirical strategy, the main enrollment results, robustness checks, and the mechanism analysis. Section 5 concludes.

[3]See De Paola and Brunello (2016) for a review of the literature.

[4]Also see Bettinger et al. (2012) and Dynarski et al. (2021) for college enrollment.

2 Background

Venezuelan Refugee Inflow in Brazil

Venezuela's political and economic collapse triggered one of the largest forced displacement crises in Latin American history. UNHCR estimates that 7.7 million Venezuelans emigrated, more than 84% of them to other countries in Latin America and the Caribbean.[5]

Figure 1: Brazil-Venezuela Border and Roraima's Municipalities

At the same time, Venezuela's education system has deteriorated significantly. Schools face severe teacher shortages and significant infrastructure problems, including power outages, lack of running water, and collapsing buildings. By 2023, 40% of Venezuelan students aged 3 to 17 attended school irregularly. In 2024, 34% of children and young adults between the ages of three and 24 (three million people) did not attend any educational institution.[6]

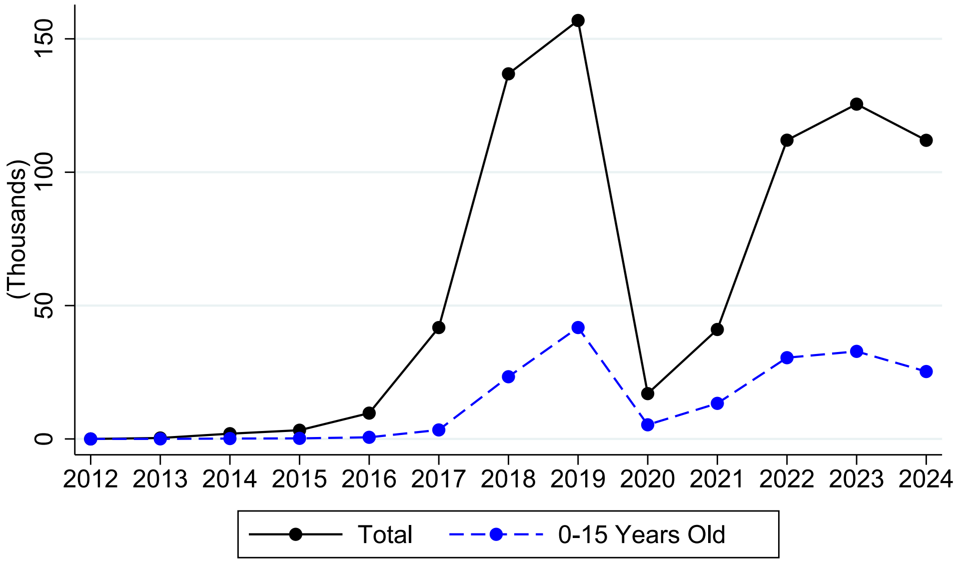
After 2017, Venezuelan migrants began arriving in large numbers at the Brazil and Venezuela border, particularly in the state of Roraima.[7] Between 2018 and 2019, nearly 400,000 Venezuelan border crossings were recorded in Roraima, accounting for more than 80% of the national inflow - see Figure 2. Within this group, inflows of children

[5] See R4V Platform for statistics by hosting country.

[6] Source: National Survey of Living Conditions (ENCOVI) Report. Also see The New Humanitarian journalistic feature.

[7] Source: Ministry of Justice and Public Security report on Venezuelan Migration for April 2024.

Figure 2: Venezuelan Net Entrance Inflow in Roraima



Source: STI.

aged 0 to 15 rose steadily from near zero in 2016 to almost 50,000 in 2019. After crossing the border at the city of Pacaraima, immigrants go to Boa Vista, Roraima's capital and biggest city (400,000 people in 2020) - see Figure 1. Once in Brazil, Venezuelan immigrants, disregarding their legal status, can access public schools.[8]

Brazilian Public Education System

Brazil's basic education system comprises 12 years of compulsory schooling (ages 6 to 18), with the academic year running from February to December. Public education is jointly funded and managed by municipal, state, and federal governments. in Boa Vista, Municipality managed schools are responsible for early childhood education (ages 0 to 3) and elementary education (ages 6 to 11). State schools serve 12 to 18 years-old students. Federal schools account for less than 2% of total enrollment. In 2019, the city had 109 public schools: 3 federal, 57 state, and 49 municipal.

To enroll a student (native or immigrant), parents or responsible adults should go to any school or a special government office, bringing their ID, proof of residence with zip code, the child's ID, and a telephone number for contact. Enrollment for new students typically takes place in January, although registration remains open throughout the

[8]Shamsuddin et al. (2021) estimates an enrollment rate of only 45% among school-age Venezuelans living in Brazil in 2020.

school year. Students who do not need or wish to change schools are automatically re-enrolled for the following year. Assignments are based on proximity to students' residences and schools' available capacity.

The Policy Change

Before September 2019, refugee children's grade assignments in Boa Vista's municipal schools were decentralized, and each school had discretion over whether and how to administer placement exams (language and math), which were often conducted in Portuguese.

In September 2019, the municipal education council enacted Resolution No. 01/2019, which standardized and centralized enrollment and grade-assignment procedures across all municipal schools. Under this new policy, placement exams were standardized, offered in Spanish, and administered by the municipal education secretariat staff in a designated location and time. An exam registry was also introduced, requiring refugee families to either call a designated phone number or visit a specific government office to place their children on a wait-list, after which they would receive a return call confirming the exam date and time.

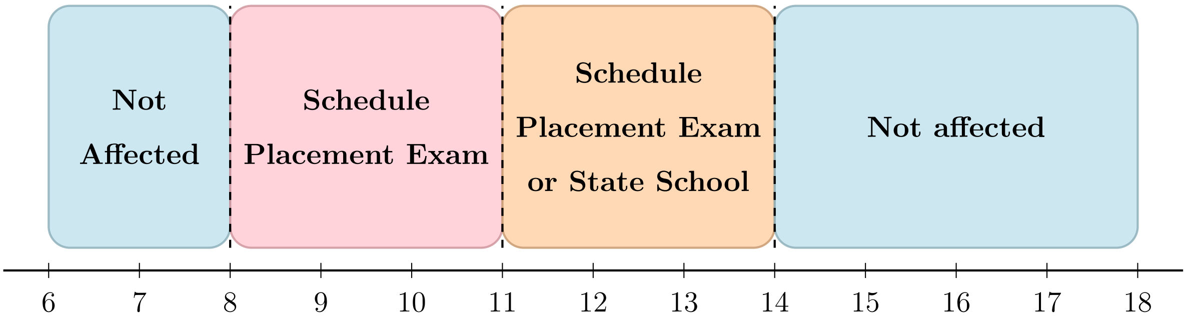
The exams followed a stepwise procedure: children were first tested on the grade level corresponding to their age—for instance, a 10-year-old took the fourth-grade exam. Those scoring below 50 percent in either language or mathematics would have to take the test corresponding to the grade below. The process was repeated until they passed or reached the first grade. All necessary exams were administered on the same day (always a weekday) at either 8 am or 2 pm.

The policy applied exclusively to Venezuelan children aged 8 to 14, as of March 31st of the enrollment year, who lacked school records or transcripts verifying their previous grade level. Importantly, it covered only municipal schools, which, in Boa Vista, offer grades 1 through 5. State-managed schools, which offer grades six and above, were not subject to the placement exam policy, and their enrollment and grade assignment procedures remained decentralized and school-specific, but generally followed students' ages. Therefore, students aged 12 or older could be admitted to state-managed schools.

Consequently, the policy design generated three differently affected subgroups.

First, children younger than eight or older than fourteen were not required to take a placement exam. Second, those aged 12 to 14 were only partially affected, as they were already eligible, based on age, for enrollment in the 6th grade and could bypass the municipal placement exam by enrolling directly in a state school. Finally, Venezuelan children aged 8 to 12 were fully subject to the policy, with no alternative non-placement exam schools to enroll in - see Figure 3.

Figure 3: Placement Exam Policy Guideline by Age Group



Between 2023 and 2024, 3,611 Venezuelan children aged 8–14 were newly enrolled in Boa Vista public schools. Over the same period, 1,913 took the placement exam, of whom 1,740 (91%) are matched to enrollment records. [9] Therefore, around 53% of enrollments occurred via the placement exam.

According to exam registry data from 2020 to 2024, children aged 12 to 14 (partially affected) account for less than 14% of total registrations (see Table 1). Moreover, age-based grade assignment was applied consistently to more than 98% of children, for example almost 99% of the 8-year-old children took the 2nd grade exam.

[9] Matching using birth dates, and expected year of enrollment based on when the exam was taken. See Table 9 in the Appendix to see statistics by age.

Table 1: Exam Registry, Take-up, and Enrollment by Age

Age	**In the Registry (Cum. %)**	**Exam Take-up Rate**	**Enrollment After Exam Take-up**
8	1,063 (22%)	68%	91%
9	1,067 (43%)	67%	91%
10	1,211 (68%)	75%	92%
11	915 (86%)	75%	95%
12	415 (95%)	66%	95%
13	179 (98%)	65%	77%
14	94 (100%)	51%	23%

Notes: Registry counts use data from 2020, 2022, 2023,and 2024. Exam take-up rates and Enrollment rates are computed for a subsample of children observed in 2023–2024. Enrollment is inferred by matching exam takers to newly enrolled Venezuelan students with the same birth date in the expected year of enrollment based on the date of the exam in the administrative data. *Source:* administrative placement exam registry (Boa Vista´s Secretary of Education) and the administrative national enrollment panel (INEP).

3 Data

National Administrative Schools' Enrollment Panel

The national enrollment panel, compiled by the National Institute for Educational Studies and Research (INEP), includes student-level identifiers and provides information on date of birth, gender, race, and nationality, with a reference date at the end of May. We define newly enrolled students in year t as those who had not appeared in the panel prior to year t. The data span the period from 2010 to 2024. We also use the *Annual School Census* to obtain school-level information on grades offered and the type of administrative management (private, state, municipal, or federal).

Boa Vista's Administrative Placement Exam Registry

Boa Vista's secretary of education keeps the list of all Venezuelan children in the exam registry since 2020. It contains the registry date, child's date of birth, the name of the child and their parent or responsible adult, the phone number left to receive the return call, and the date and time of the exam. To identify the gender of students, we combine different list of the most common names and associated gender among Venezuelan immigrants or general population in Brazil, Colombia and United States.

[10] Finally, for 2023 and 2024, we also observe whether children were living in a refugee shelter and whether they took the exam.

Asylum Application Data

To capture the demographic composition of Venezuelan refugees over time, we use asylum application data from the *Comitê Nacional para os Refugiados* (CONARE), the collegiate body within the Ministry of Justice and Public Security responsible for adjudicating refugee status claims in Brazil. This dataset provides detailed individual information on asylum cases, including nationality, year of birth, municipality where the application was filed, and reported year of entry.

4 Empirical Strategy

We first graphically compare the age distribution of newly enrolled Venezuelan students in Boa Vista public schools before and after the policy implementation (September 2019). We grouped all enrollments from 2017 (when the inflow of school-age Venezuelans started) to 2019, and 2020 to 2024 - Figure 4a. Between 2017 and 2019, the enrollment share smoothly decreased along with Venezuelans' age. After 2019, however, the distribution exhibits a pronounced "missing mass" between ages 8 and 11, leading to relatively higher shares among the remaining age groups of enrolled Venezuelans.[11]

Figure 4b displays the yearly age distribution of newly enrolled Venezuelan students. Because the enrollment data consist of repeated cross-sections with a reference month of May each year, the 2020 cohort reflects a mix of students enrolled before and after the introduction of the placement exam requirement and thus captures the transition into the policy. Consistent with this, 2020 is the only post-2019 year in which the "gap" is not yet fully pronounced.

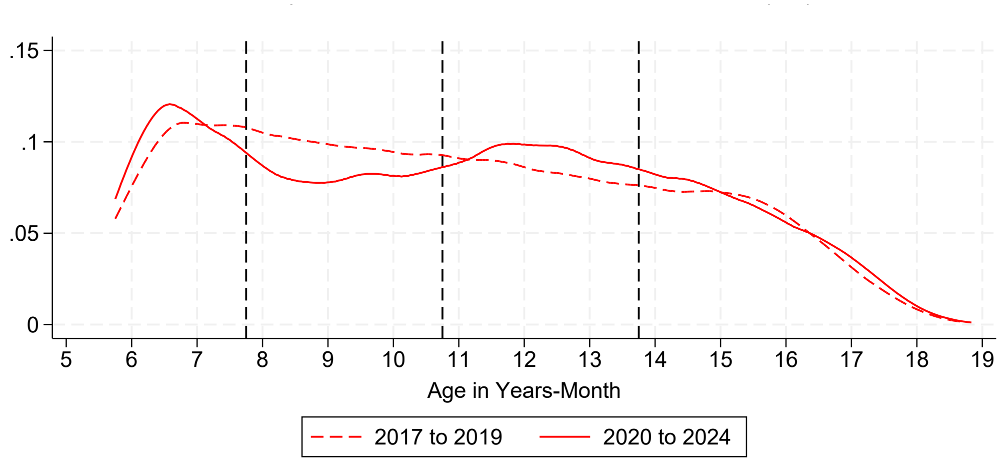
[10] Brazil: Roraima's most common names from the 2022 IBGE Census. Colombia: Venezuelan Immigrants most common names from the 2024 INE Census. United States: most common names in high Venezuelan immigration states (FL, TX, IL, GA, NY) from the Social Security Administration Top 1000 names.

[11] When separating enrollment by gender, we also observe similar changes in the PDF for boys and girls - see Figure 19 in the Appendix.

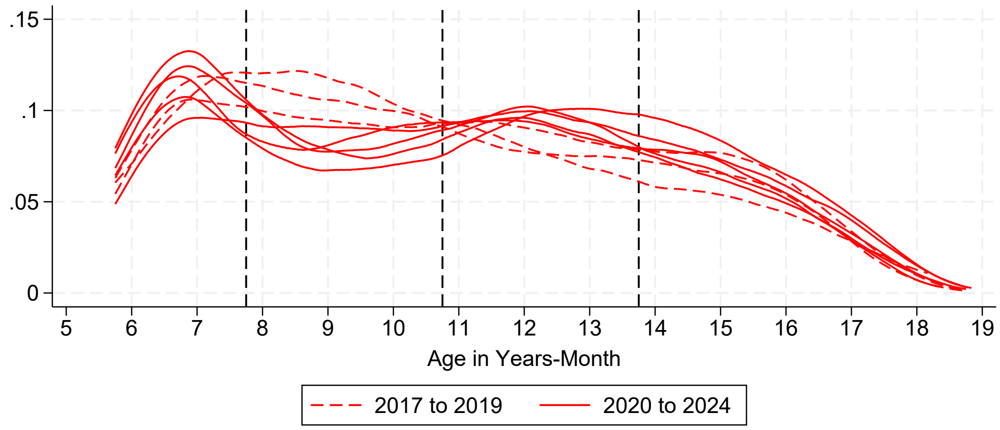
Moreover, no such gap is observed in the three years prior to the policy, suggesting that the shift in the age distribution is not driven by pre-existing demographic changes among Venezuelan students. Finally, it is unlikely that the pandemic explains this pattern, as the distributions for 2022, 2023, and 2024 exhibit similar features.

Figure 4: Age PDF of Newly Enrolled Venezuelans - Boa Vista

(a) Before vs After Policy



(b) Before vs After Policy (Yearly)

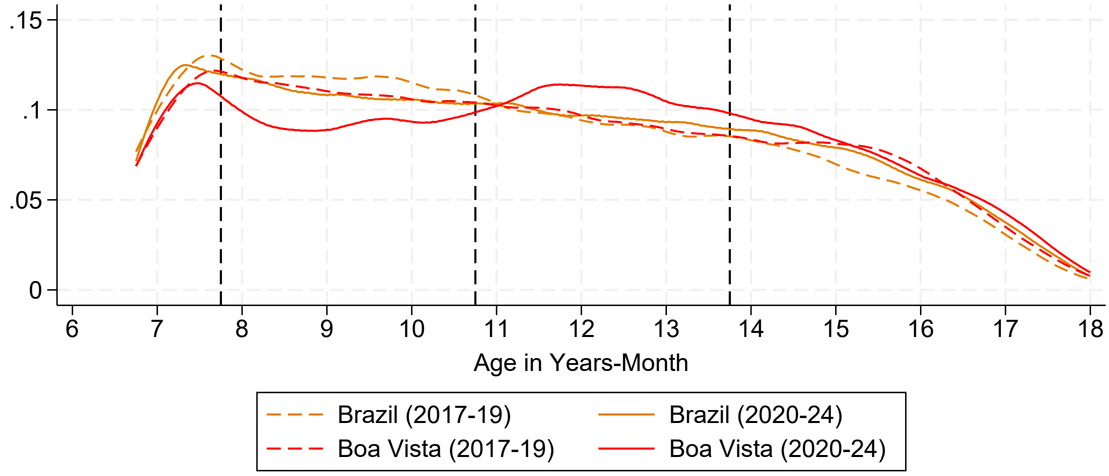


Notes: enrollment across all public schools in Boa Vista. Age in years-month defined based on the reference date of March 31st. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

We can also compare the enrollment patterns of Venezuelans in Boa Vista with those in the rest of the country, where no such placement exam requirement was implemented. According to Figure 5, the age distribution of Venezuelans enrolled in public schools in the rest of the country had a similar shape to that observed in Boa

Vista before the policy.[12] After 2019, only Boa Vista experienced such a change in enrollment age distribution.

Figure 5: Age PDF of Newly Enrolled Venezuelans (Yearly) - Boa Vista Vs Brazil



Notes: enrollment across all public schools. *"Brazil"* category excludes Roraima state. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

This setting allows us to explore different specifications to identify the policy's causal effect on enrollment. First, we estimate a Diff-in-Diff using the following TWFE specification:

$$Y_{at} = \alpha + \beta_1 \, \text{Post}_t + \beta_2 \, \text{Treated}_a + \beta_3 \, (\text{Post}_t \times \text{Treated}_a) + FE + \varepsilon_{at} \tag{1}$$

Y_{at} is the number of Venezuelans just enrolled in Boa Vista public schools in age group a (age in months) in year t. $Post_t$ is a dummy after 2019. Treated_a is a dummy for ages 8 to 14. The parameter of interest is β_3 and its causal interpretation holds as long as there were no demographic shifts among Venezuelan migrants in Boa Vista after 2019 within the affected age groups.

Differently, we can also estimate a Diff-in-Diff-in-Diff specification, exploring the targeted age groups and the geographic concentration of the policy (Boa Vista vs. the rest of Brazil):

[12] See Figure 20 in the Appendix for the yearly age density among newly enrolled Venezuelans in the rest of the country.

$$
\begin{aligned}
Y_{art} = \alpha + \beta_1 \, \text{Post}_t + \beta_2 \, \text{BV}_r + \beta_3 \, \text{Treated}_a + \\
& \beta_4 \, (\text{Post}_t \times \text{BV}_r) + \beta_5 \, (\text{Post}_t \times \text{Treated}_a) + \beta_6 \, (\text{BV}_r \times \text{Treated}_a) + \\
& \beta_7 \, (\text{Post}_t \times \text{BV}_r \times \text{Treated}_a) + FE + \varepsilon_{art}
\end{aligned} \tag{2}
$$

Y_{art} is the number of Venezuelans enrolled in public schools in age group a (age in months) in region r (either Boa Vista or the rest of the country) in year t. $Post_t$ is a dummy after 2019. BV_r is a dummy for Boa Vista and Treated_a is a dummy for ages 8 to 14. The parameter of interest is β_7. We explored different FE, including time, age-region, age-time, and region-time. The causal interpretation of β_7 remains valid even in the presence of demographic changes among Venezuelan migrants, provided that such changes are common to both Boa Vista and the rest of Brazil.

Finally, we can also estimate another Diff-in-Diff-in-Diff specification, exploring the targeted age groups and the nationality of the affected group (Venezuelans vs. Brazilians):

$$
\begin{aligned}
T_{ant} = \alpha + \beta_1 \, \text{Post}_t + \beta_2 \, \text{Ven}_n + \beta_3 \, \text{Treated}_a + \\
& \beta_4 \, (\text{Post}_t \times \text{Ven}_n) + \beta_5 \, (\text{Post}_t \times \text{Treated}_a) + \beta_6 \, (\text{Ven}_n \times \text{Treated}_a) + \\
& \beta_7 \, (\text{Post}_t \times \text{Ven}_n \times \text{Treated}_a) + FE + \varepsilon_{ant}
\end{aligned} \tag{3}
$$

T_{ant} is the total number of students from nationality n (Brazilians or Venezuelans) enrolled in Boa Vista public schools in age group a (age in months) in year t. $Post_t$ is a dummy after 2019. Ven_n is a dummy for Venezuelan nationality and Treated_a is a dummy for ages 8 to 14. The parameter of interest is β_7. We explored different FE, including time, age-nationality, age-time, and nationality-time. Using Brazilians as the control group, capture shocks to the Boa Vista educational system, such as capacity of schools offering grades for affected age-groups.

None of the empirical strategies above can account for Boa Vista–specific demographic shifts among Venezuelan migrants in the affected age groups. To address this concern, Section 6 shows that no such shock is observed when examining the age distribution of asylum applications in Boa Vista.

5 Results

Table 2 summarizes the estimates from Equations 1, 2, and 3. The results are consistent across specifications and indicate that the placement exam policy reduced Venezuelan enrollment in the target age group (8 to 14) by 10% to 17%. This numbers correspond to 180 to 306 fewer enrollments per year among Venezuelans ages 8 to 14.

These findings are robust to alternative estimators, including PPML and OLS with the outcome expressed in logs. Moreover, the consistency of the triple interaction estimate across both panels B and C, despite their reliance on different control groups, is unlikely to arise from confounding factors that would simultaneously bias both in the same direction and by a similar magnitude.

We note that the coefficient on $\text{Post}_t \times \text{BV}_r$ in Panel B is large and statistically significant, indicating that Venezuelan enrollment in Boa Vista grew considerably less than in the rest of Brazil after 2019 across all age groups. This pattern most likely reflects the differential timing and scale of the Venezuelan settlement across Brazil. Boa Vista served as the primary entrance for Venezuelans in the country, and consequently experienced earlier and larger shock to enrollment. By 2019, enrollment growth in Boa Vista was decelerating, while Venezuelan enrollment in the rest of the country was still accelerating from a much smaller base.

Finally, Figure 6 provides direct evidence in support of the parallel trends assumption underlying our difference-in-differences designs. In the three years prior to the policy's implementation, the estimated year-specific coefficients are small, statistically indistinguishable from zero, and show no systematic trend, indicating that enrollment trajectories for affected and unaffected age groups were evolving in parallel before the policy was implemented. In addition, the effects do not fade over time indicating that the policy's enrollment effect was sustained throughout the post-period and that families adaption to the new requirements was unlikely or not relevant (what we confirm in the mechanism analysis).

Table 2: Placement Exam Impact on Venezuelan Enrollment

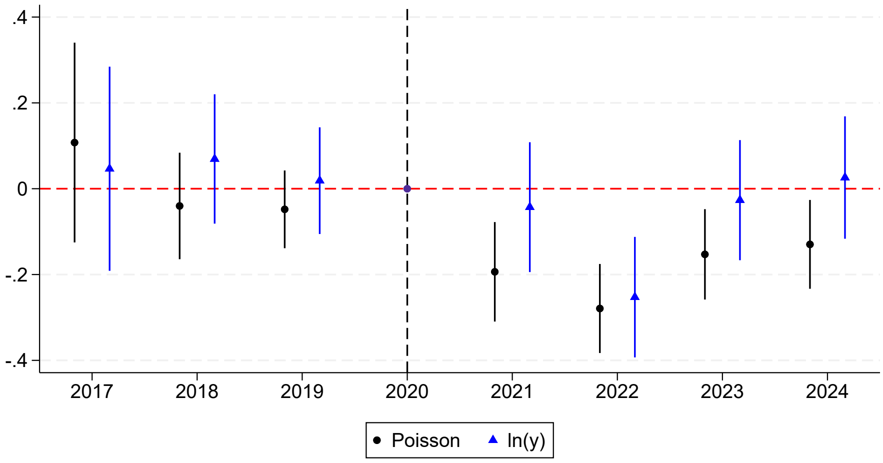
Panel A: DD (Eq. 1)	*Poisson*	*ln(Y_{at})*
Post$_t$ × Treated$_a$	**-0.107***** **(0.030)**	**-0.104**** **(0.046)**
Observations	1,264	1,188
R^2	0.626	0.887
Mean(Y_{at})	19.01	20.23
Time FE	Y	Y
Age FE	Y	Y

Panel B: DDD (Eq. 2)	*Poisson*		*ln(Y_{art})*	
Post$_t$ × BV$_r$ × Treated$_a$	**-0.054** **(0.052)**	**-0.077***** **(0.028)**	**-0.154**** **(0.074)**	**-0.143**** **(0.067)**
Post$_t$ × BV$_r$	-1.399*** (0.037)		-1.599*** (0.056)	
Post$_t$ × Treated$_a$	-0.053 (0.033)		0.044 (0.054)	
Observations	2,536	2,460	2,399	2,399
R^2	0.867	0.890	0.927	0.978
Mean(Y_{art})	39.92	41.15	42.2	42.2
Time FE	Y	N	Y	N
Region-age FE	Y	Y	Y	Y
Age-time FE	N	Y	N	Y
Region-time FE	N	Y	N	Y

Panel C: DDD (Eq. 3)	*Poisson*		*ln(T_{ant})*	
Post$_t$ × Ven$_n$ × Treated$_a$	**-0.051** **(0.080)**	**-0.074***** **(0.025)**	**-0.150*** **(0.088)**	**-0.180***** **(0.049)**
Post$_t$ × Ven$_n$	1.280*** (0.059)		1.840*** (0.062)	
Post$_t$ × Treated$_a$	-0.024*** (0.009)		-0.013 (0.052)	
Observations	2,544	2,544	2,506	2,506
R^2	0.935	0.958	0.929	0.990
Mean(T_{ant})	205.3	205.3	208.4	208.4
Time FE	Y	N	Y	N
Nationality-age FE	Y	Y	Y	Y
Age-time FE	N	Y	N	Y
Nationality-time FE	N	Y	N	Y

Notes: Pseudo-R^2 reported for Poisson estimations. Robust standard errors are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Figure 6: Event Study Results (Eq. 1)



To evaluate differential policy effects across affected ages, we estimate heterogeneous affect for Eq. 1, and estimate equations 2 and 3 using two samples, including the control ages and either the 8 to 11-year-olds or the 12 to 14-year-olds (who had the option to enroll in non-placement exam state-managed schools). As expected, the policy had no impact on the later group and decreased enrollment among highly affected age groups between 16% to 33% - see Table 3.

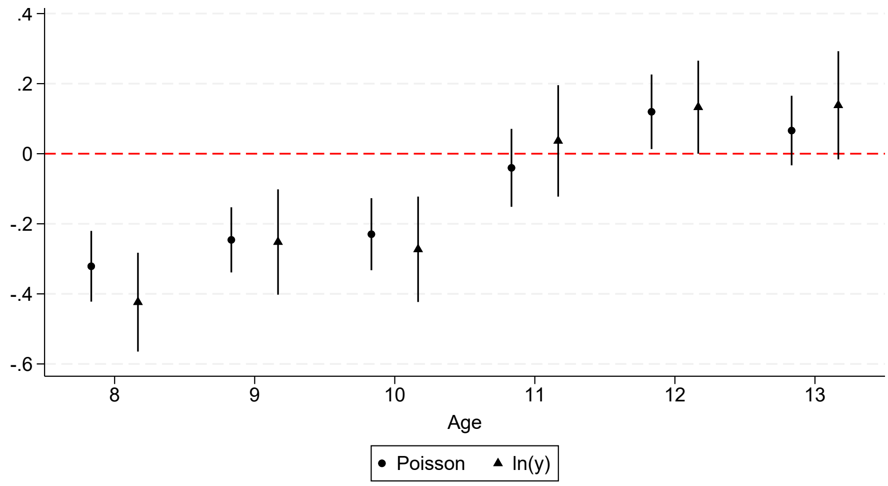
Figure 7 reinforces this pattern: the negative treatment effect is concentrated at ages 8 to 11 and transitions toward zero at age 12, precisely tracking the boundary between municipal and state school eligibility. This age-gradient in the treatment effect, mirroring the institutional discontinuity in the policy design, is unlikely to arise from any confounding factor that does not share the same pattern providing an additional layer of causal identification.

Table 3: Heterogeneous Effect on Enrollment by Age Group

Panel A: DD (Eq. 1)	Poisson		$\ln(Y_{at})$	
	(1)	(2)	(3)	(4)
$\text{Post}_t \times \text{Treated}_a$	0.048	0.048	0.103*	0.103*
	(0.036)	(0.036)	(0.053)	(0.053)
$\text{Post}_t \times \text{Treated}_a \times \text{8-11}_a$	**-0.314*** **	**-0.314*** **	**-0.419*** **	**-0.419*** **
	(0.039)	**(0.039)**	**(0.055)**	**(0.055)**
Observations	1,264	1,264	1,188	1,188
R^2	0.629	0.629	0.892	0.892
$\text{Mean}(Y_{at})$	19.01	19.01	20.23	20.23
Time FE	Y	Y	Y	Y
Age FE	Y	Y	Y	Y
Panel B: DDD (Eq. 2)	**8-11 Years Old**		12-14 Years Old	
	Poisson	$\ln(Y_{art})$	Poisson	$\ln(Y_{art})$
$\text{Post}_t \times \text{BV}_r \times \text{Treated}_a$	**-0.167*** **	**-0.310*** **	0.024	0.020
	(0.033)	**(0.079)**	(0.033)	(0.082)
Observations	1,872	1,818	1,886	1,830
R^2	0.895	0.978	0.891	0.978
$\text{Mean}(Y_{art})$	39.85	41.03	38.72	39.9
Time FE	N	N	N	N
Region-age FE	Y	Y	Y	Y
Age-time FE	Y	Y	Y	Y
Region-time FE	Y	Y	Y	Y
Panel C: DDD (Eq. 3)	**8-11 Years Old**		12-14 Years Old	
	Poisson	$\ln(T_{ant})$	Poisson	$\ln(T_{ant})$
$\text{Post}_t \times \text{Ven}_n \times \text{Treated}_a$	**-0.190*** **	**-0.331*** **	0.048	-0.030
	(0.028)	**(0.056)**	(0.029)	(0.056)
Observations	1,952	1,917	1,968	1,931
R^2	0.959	0.99	0.96	0.99
$\text{Mean}(T_{ant})$	197.8	201.4	199.3	203.1
Time FE	N	N	N	N
Region-age FE	Y	Y	Y	Y
Age-time FE	Y	Y	Y	Y
Region-time FE	Y	Y	Y	Y

Notes: Pseudo-R^2 reported for Poisson estimations. Robust standard errors are reported in parentheses.
*** p<0.01, ** p<0.05, * p<0.1.

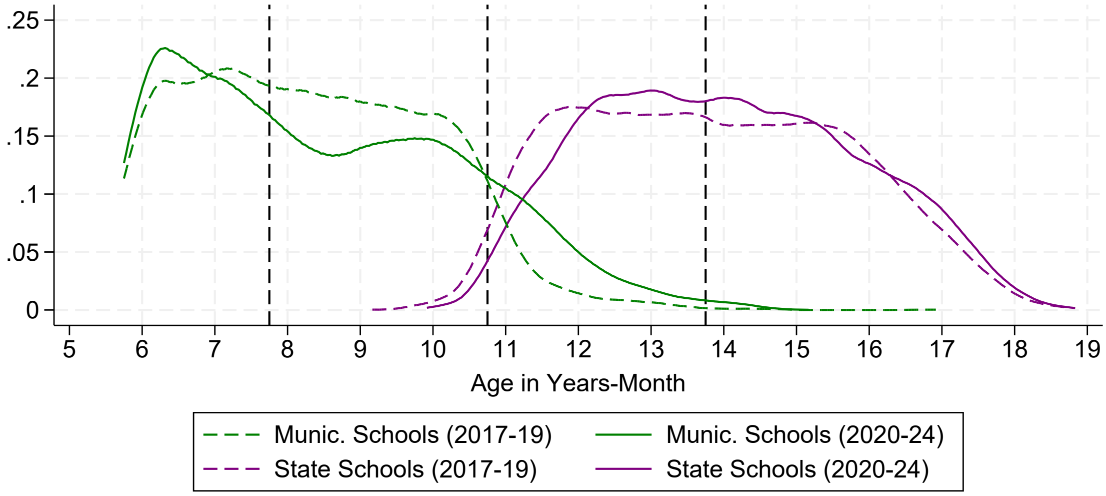
Figure 7: Results by Age (Eq. 1)



Considering that state and municipality schools cover different student-age populations and the exam requirement affected only the municipality school enrollment process, we next examine heterogeneous effects by school management type.

First, Figure 8 shows that municipal schools primarily serve grades 1 to 5 (roughly ages 6 to 11), while state schools cover grades 6 to 12 (ages 12 and above). After 2019, the distribution for municipal schools exhibits a pronounced deficit at ages 8 to 11 and a modest increase in municipal school enrollment at ages 11 to 13 that are not typically served by those schools. This pattern is mirrored by a decline in the share of students 11-13 within state schools. Together, these shifts suggest that the policy's implementation and outreach may have redirected students across the two school systems, which can be translated into higher age-distortion rates among 5th graders - confirmed and discussed in section 7.

Figure 8: Age of Newly Enrolled Venezuelans - by School Type



Notes: PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

To formally verify the differential policy effects by school type, we explore a school-level panel to estimate the following difference-in-differences using either state managed schools or municipality managed schools:

$$Y_{ast} = \alpha + \beta_1 \, \text{Post}_t + \beta_2 \, \text{Treated}_a + \beta_3 \, (\text{Post}_t \times \text{Treated}_a) + FE + \varepsilon_{ast} \tag{4}$$

The results by school type, reported in Table 4, confirm the graphic patters of Figure 8. The coefficient on $\text{Post}_t \times \text{Treated}_a \times 8\text{-}11_a$ in Panel A is negative, large, and highly significant for municipal schools, while the corresponding estimate in Panel B for state-managed schools is small and statistically indistinguishable from zero across all specifications. Moreover, the Poisson estimates confirm an increase in the enrollment of newly enrolled Venezuelans aged 11 to 14 in municipal schools, indicating that these schools absorbed students they do not typically serve after the policy was implemented.

Table 4: Policy's Enrollment Effect by School Type

DD (Eq. 4)	**Municipality Managed Schools**		**State Managed Schools**	
	Poisson	$\ln(Y_{ast})$	Poisson	$\ln(Y_{ast})$
$\text{Post}_t \times \text{Treated}_a$	0.768***	0.052	0.054	0.033
	(0.075)	(0.042)	(0.052)	(0.034)
$\text{Post}_t \times \text{Treated}_a \times \text{8-11}_a$	-1.058***	-0.174***		
	(0.073)	(0.040)		
Observations	24,320	8,259	18,896	5,581
R^2	0.203	0.399	0.200	0.474
$\text{Mean}(Y_{ast})$	0.505	1.488	0.419	1.42
Time FE	Y	Y	Y	Y
School-age FE	Y	Y	Y	Y

Notes: Pseudo-R^2 reported for Poisson estimations. Robust standard errors are reported in parentheses.
*** p<0.01, ** p<0.05, * p<0.1.

6 Robustness

Spillover to other Municipalities

Potential domestic migration of affected Venezuelan students could lead to an overestimation of the effects in Equation 1. In addition, a geographic spillover analysis helps assess whether the policy genuinely affected enrollment rates or simply geographically redistributed affected Venezuelan children.

To formally test those spillovers, we estimate Eq. 1 using data from different regions in the country representing potential destination of affected Venezuelans. The selection of those regions is guided by the geography of Venezuelan settlement in Brazil and the relocation options available to families residing in Boa Vista. We focus on three destinations: the rest of Brazil, the rest of Roraima excluding Boa Vista, and Manaus. The rest of Roraima is a natural destination due to its proximity and direct road connection to Boa Vista. Manaus is the nearest large urban center (with a population exceeding two million) and the second-largest recipient of enrolled Venezuelan students in the country.

According to the results in Table 5, we do not observe any increase in enrollment in this age group in those destinations. The point estimates for the triple interaction term are small, statistically insignificant, and in several specifications take the wrong sign for a domestic migration story. This pattern is consistent with genuine enrollment suppression.

Table 5: Policy's Enrollment Effect Spillovers

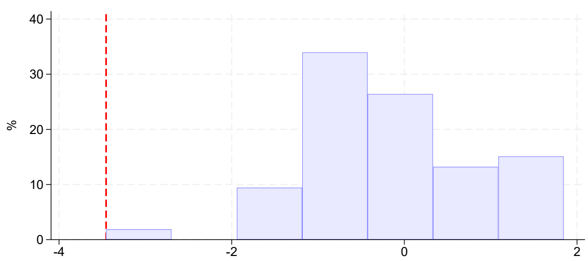
DD - Eq. 1	**Rest of Brazil**		**Rest of Roraima**		**Manaus**	
	Poisson	$ln(Y_{at})$	*Poisson*	$ln(Y_{at})$	*Poisson*	$ln(Y_{at})$
$\text{Post}_t \times \text{Treated}_a$	0.003	0.07	0.076	0.089	-0.051	0.009
	(0.033)	(0.059)	(0.074)	(0.084)	(0.051)	(0.056)
$\text{Post}_t \times \text{Treated}_a \times \text{8-11}_a$	-0.104**	0.073	-0.087	-0.056	0.009	0.090
	(0.038)	(0.062)	(0.078)	(0.093)	(0.056)	(0.071)
Observations	1,272	1,211	1,256	1,007	1,272	1,102
R^2	0.898	0.959	0.456	0.700	0.681	0.884
Mean(Y)	60.69	63.75	4.81	6.00	11.27	13.00
Time FE	Y	Y	Y	Y	Y	Y
Age FE	Y	Y	Y	Y	Y	Y

Notes: Robust standard errors are reported in parentheses.
*** p<0.01, ** p<0.05, * p<0.1.

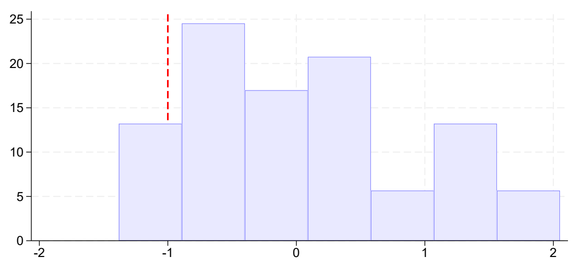
To further verify for geographic spillovers and strengthen the causal interpretation of the results, we implement a simulation exercise. First, we identified the top 10% of municipalities in the country receiving Venezuelan students between 2017 and 2019, corresponding to a total of 53 municipalities, with Boa Vista ranking first. Next, we iteratively select each of these cities as a placebo-treated unit and estimate Equations 1 and 2. Finally, we construct the distribution of the resulting estimated effects and corresponding t-statistics, and compare them to the estimates for Boa Vista reported in Table 2.

Figure 9: Placebo-Treated Municipalities Simulation - T-stat. Distribution

(a) Poisson

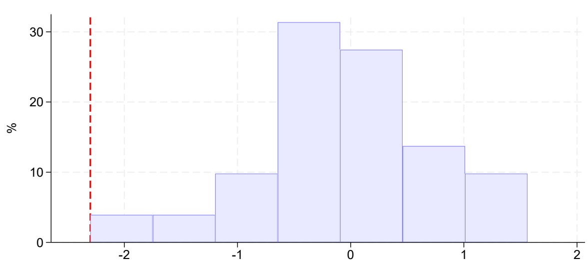


Eq. 1 - $\boldsymbol{\beta_3}$ (Post$_t$ $\times$ Treated$_a$)

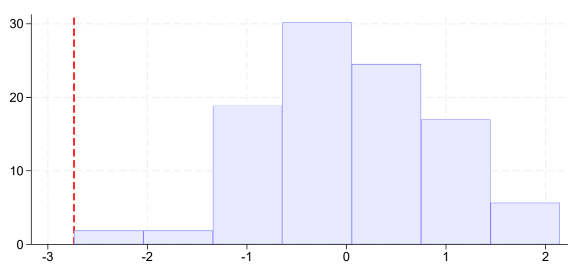


Eq. 2 - $\boldsymbol{\beta_7}$ (Post$_t$ $\times$ BV$_r$ $\times$ Treated$_a$)

(b) ln(Y)



Eq. 1 - $\boldsymbol{\beta_3}$ (Post$_t$ $\times$ Treated$_a$)



Eq. 2 - $\boldsymbol{\beta_7}$ (Post$_t$ $\times$ BV$_r$ $\times$ Treated$_a$)

Notes: The vertical dashed line represents the estimates for Boa Vista, reported in Table 2. Percentiles of the t-statistics are as follows: for β_3 estimated using a Poisson specification, 0; for β_3 estimated using $\ln(Y)$, 0; for β_7 estimated using a Poisson specification, 11.3%; and for β_7 estimated using $\ln(Y)$, 1.9%.

According to Figure 9, the distribution of placebo estimates is approximately centered at zero for both the DiD (Equation 1) and DDD (Equation 2) specifications. Most importantly, Boa Vista's t-statistic sits in the extreme left tail of the placebo distribution in all four panels. For the DiD specification estimated via Poisson and $\ln(Y)$, Boa Vista's estimate is at the 0th percentile of the distribution, therefore, no other municipality among the top 10% of Venezuelan-receiving cities produces an estimate as negative as Boa Vista's. For the DDD specification, Boa Vista's estimate falls at the 11.3th and 1.9th percentiles of the Poisson and $\ln(Y)$ placebo distributions respectively.

Finally, the share of placebo municipalities producing a positive and statistically significant enrollment effect is negligible across all four panels. This pattern confirms the absence of geographic positive spillover effects in enrollment among other municipalities of the country.

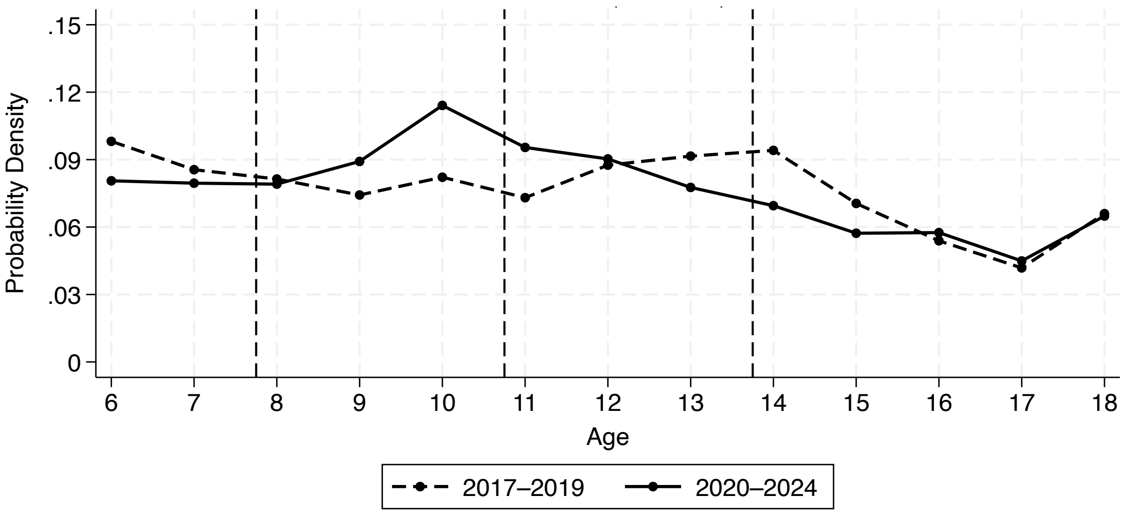
Boa Vista's Refugee Population Demographic Shocks

The data so far explored covers only Venezuelans enrolled. Therefore, demographic changes specific to Boa Vista and the 8 to 14 years old Venezuelan population residing in the city could be driving the results. To verify this we explore the CONARE data on the age distribution of Venezuelan children and teenagers applying for refugee status based on their municipality of residency in the application and reported year of arrival in the country.

Crucially, filing an asylum application is typically one of the first steps toward integration upon arrival and a key immediate action reinforced by the government reception policy ("Operação Acolhida"). The CONARE data, therefore, provides a measure of the school-age Venezuelan population in Boa Vista.

From Figure 10, we don't observe graphically any significant decline in the number of Venezuelan refugees within the targeted age group after the policy implementation.

Figure 10: Age Distribution of Venezuelan Asylum Cases - Boa Vista



Notes: Sample restricted to Venezuelan children aged 6–18 with residency in Boa Vista (RR). Year based on reported entrance year into the country. Age calculated as of January 1st of the year. *Source:* CONARE.

To formally test for demographic changes in the school-age Venezuelan refugee population, we estimate equations 1 and 2 using the total number of applications as an outcome. Given that the data only provides the year of birth of applicants, age is measured in years.

According to the results reported in Table 6, we find no evidence of a differential decline in asylum applications among children aged 8 to 14 in Boa Vista after 2019 relative to other age groups or relative to the rest of Brazil. If anything, the point estimates for the affected age group are mostly positive, suggesting an increase in applications among children in the placement exam targeted age range. The enrollment decline we document therefore reflects a genuine decrease in enrollment rates among affected children, rather than a mechanical consequence of demographic change in the underlying population in Boa Vista.

Table 6: Placement Exam Impact on Venezuelan Asylum Cases

Panel A: DD (Eq. 1)	*Poisson*		$ln(Applications_{at})$	
Post$_t$ × **Treated**$_a$	0.223***	0.159*	0.138	0.121
	(0.068)	(0.083)	(0.083)	(0.107)
Post$_t$ × **Treated**$_a$ × **8-11**$_a$		0.128		0.035
		(0.106)		(0.127)
Observations	104	104	104	104
R^2	0.866	0.867	0.947	0.947
Mean(Y)	209.1	209.1	209.1	209.1
Time FE	Y	Y	Y	Y
Age FE	Y	Y	Y	Y

Panel B: DDD (Eq. 2)	All		8-11 Years Old	
	Poisson	$ln(Applications_{art})$	Poisson	$ln(Applications_{art})$
Post$_t$ × **BV**$_r$ × **Treated**$_a$	-0.001	-0.033***	0.062***	0.030***
	(0.021)	(0.000)	(0.016)	(0.000)
Observations	208	208	176	176
R^2	0.963	0.989	0.963	0.990
Mean (Y)	303.4	303.4	299.9	299.9
Time FE	N	N	N	N
Region-age FE	Y	Y	Y	Y
Age-time FE	Y	Y	Y	Y
Region-time FE	Y	Y	Y	Y

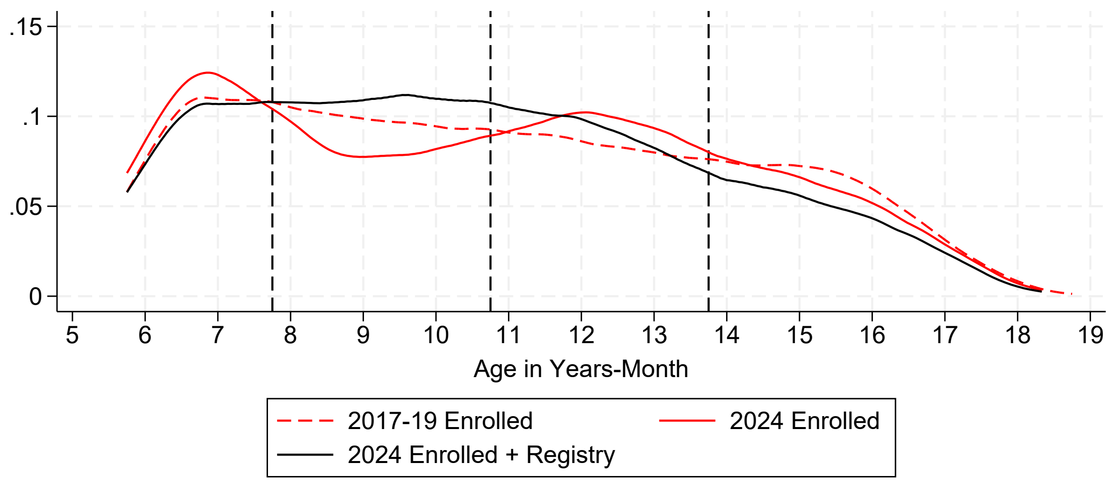
Notes: Pseudo-R^2 reported for Poisson estimations. Robust standard errors are reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

7 Mechanisms and Age-Distortion Consequences

What is the main barrier?

In this section, we examine the main mechanisms underlying the decline in enrollment. The placement exam process involves two key steps: registering children on a waitlist and attending the exam. Accordingly, the negative enrollment effect may arise either from families not registering their children or from children failing to attend the exam after registration.

Figure 11: Age Density of Newly Enrolled Venezuelans



Notes: Enrolled + Registry includes Venezuelan students enrolled and the ones in the registry who didn't take the exam or who took the exam and were not matched using birth date and enrollment/exam year in the enrollment data.

Figure 11 overlays three age density curves for newly enrolled Venezuelan children in Boa Vista: the pre-policy distribution (2017–2019), the post-policy enrollment distribution (2024), and a post-policy distribution that adds children on the exam registry who did not ultimately enroll (either because they didn't take the exam or because they were not matched in the enrollment panel).[13] The third curve (*Enrollment + Registry*) close the gap observed among affected ages. This visual comparison indicates that the main barrier to enrollment imposed by the placement exam was not in families' willingness to register children.

To formalize this comparison, we re-estimates the main specifications using enroll-

[13] We only have data on exam attendance for 2023 and 2024

ment plus registry as the outcome variable. The triple interaction term is now positive and statistically significant across all specifications, indicating that once registered children are counted alongside enrolled children, the post-policy deficit in the affected age group disappears and is in fact reversed. The positive coefficients is consistent with the potential increase in the population of affected age children observed in the CONARE data. Moreover, the registry potentially includes families that would not enroll their children even in the absence of the policy, for example, transit families that planned to move to other parts of the country.

Table 7: Placement Exam Effect on *Enrollment + Registry*

Panel A: DD (Eq. 1)	*Poisson*	$ln(Y_{at} + R_{at})$
$\text{Post}_t \times \text{Treated}_a$	0.209***	0.298***
	(0.045)	(0.073)
$\text{Post}_t \times \text{Treated}_a \times \text{8-11}_a$	0.025	0.056
	(0.045)	(0.071)
Observations	616	575
R^2	0.713	0.916
Mean(Y_{at})	17.92	19.20
Time FE	Y	Y
Age FE	Y	Y

Panel B: DDD (Eq. 2)	8-11 Years Old		12-14 Years Old	
	Poisson	$ln(Y_{art} + R_{art})$	*Poisson*	$ln(Y_{art} + R_{art})$
$\text{Post}_t \times \text{BV}_r \times \text{Treated}_a$	0.396***	0.346***	0.208***	0.269*
	(0.038)	(0.131)	(0.039)	(0.136)
Observations	896	869	902	873
R^2	0.907	0.978	0.906	0.978
Mean(Y_{art})	31.45	32.45	29.94	30.95
Time FE	N	N	N	N
Region-age FE	Y	Y	Y	Y
Age-time FE	Y	Y	Y	Y
Region-time FE	Y	Y	Y	Y

Notes: Pseudo-R^2 reported for Poisson estimations. Robust standard errors are reported in parentheses.
*** p<0.01, ** p<0.05, * p<0.1.

Between 2023 and 2024, approximately 3,300 children were registered on the placement exam wait list in Boa Vista. Of these, 30% did not attend the exam. Next we verify the correlations behind exam absenteeism by estimating equation 5. Our main explanatory variables measure the extent and quality of communication with the families to inform the dates of exams and how long the family is expected to wait for their children to take the exam.

$$\begin{aligned} \textit{Missed Exam}_i = \alpha &+ \beta_1 \, \text{Wait List Position}_i \\ &+ \beta_2 \, \text{Phone}_i + \beta_3 \, \text{Multiple Phones}_i + \beta_4 \, \text{NGO Phone}_i \\ &+ FE + \varepsilon_i. \end{aligned} \tag{5}$$

The dependent variable is *Missed Exam*$_i$, a dummy for student "i" not taking the exam. Phone$_i$ is a dummy equal to one if the individual provided at least one valid phone number at registration, and Multiple Phones$_i$ indicates whether more than one phone number was annotated. NGO Phone$_i$ is a dummy equal to one if one of the contact numbers belongs to a humanitarian organization. Around 8% of registered families rely on NGO phone numbers that can operate as an intermediary agent in the communication.

The time waited between registering the children and taking the exam is correlated with how easy it is to contact the family (exams can be postponed until communication is complete). Therefore, to measure how the capacity/speed of the exam scheduling process affects absenteeism, we instead use the variable Wait List Position$_i$, which measures how far down the queue the individual was when registered for the exam. Higher positions reflect longer waiting times - see Figure 21 in the Appendix. This proxy for waiting time is less likely to be correlated with family and children unobservable characteristics. Finally, we explore different sets of fixed effects (FE), including year, age, and month-year of registry. Our preferred specification compares students with the same age and registered in the same month and year.

According to the results in Table 8, the higher the position in the waiting list, the higher the chances of absenteeism. In addition, as expected, the existence of a phone number in the registry and the registry of multiple phone numbers can facilitate communication and are associated with smaller absenteeism. However, having an NGO phone number is associated with higher chances of absenteeism, likely because this variable captures other socio-economic variables of student "i" and their family. This becomes clear once we add a dummy for whether children "i" live in a shelter (only available for a subsample), and the coefficient associated with NGO phone number decreases and loses significance - see column (4) of Table 8.[14]

Absenteeism is considerably higher (around 20 percentage points) for refugee children living in shelters. The addition of this variable, however, does not change the effects of wait list position and having multiple contact phone numbers, reinforcing the importance of those two logistical components in mediating exam attendance. Results also remain unchanged with the inclusion of students gender, and the non-significance of the "Girl" dummy speaks with the similar impact of the policy for girls and boys observed graphically by Figure 19 in the Appendix. Finally, the interaction of phone communication variables and position in the wait list (results not reported in this draft) is not significant, and therefore, indicates that both sets of variables matter by themselves.

[14]The results are similar when estimating equation 5 using logit - see Table 10 in the Apppendix.

Table 8: What Explains Exam Absenteeism

Eq. 5	Y = *Missed Exam*$_i$				
	(1)	(2)	(3)	(4)	(5)
Wait List Position	0.0006***	0.0006***	0.0006***	0.0009***	0.0008***
	(0.0000)	(0.0000)	(0.0001)	(0.0002)	(0.0002)
Have Phone Number	-0.1245***	-0.1268***	-0.1059**	-0.2017	-0.1838
	(0.0464)	(0.0462)	(0.0450)	(0.1352)	(0.1485)
Have 2+ Phone Numbers	-0.0958***	-0.0913***	-0.0609***	-0.0702**	-0.0904**
	(0.0194)	(0.0194)	(0.0189)	(0.0318)	(0.0365)
Have NGO Phone Number	0.1680***	0.1717***	0.1746***	0.0549	0.0624
	(0.0246)	(0.0245)	(0.0238)	(0.0375)	(0.0430)
Live in a Shelter				0.2055***	0.2218***
				(0.0300)	(0.0338)
Girl					0.0066
					(0.0261)
Observations	3,298	3,298	3,296	1,472	1,083
R-squared	0.3407	0.3479	0.4006	0.3078	0.3381
Fixed Effects:					
Year FE	Y	Y	N	N	N
Age FE	N	Y	Y	Y	Y
Month-Year Registered FE	N	N	Y	Y	Y
Avg Missed Exam	0.299	0.299	0.299	0.496	0.501
Avg. Position Wait List	222.2	222.2	222.3	325.3	330.8
Avg. Have a phone	0.980	0.980	0.980	0.993	0.993
Avg. Have 2+ phones	0.130	0.130	0.130	0.147	0.146
Avg. Have NGO phone	0.0797	0.0797	0.0798	0.147	0.141
Avg. Live in Shelter				0.357	0.355

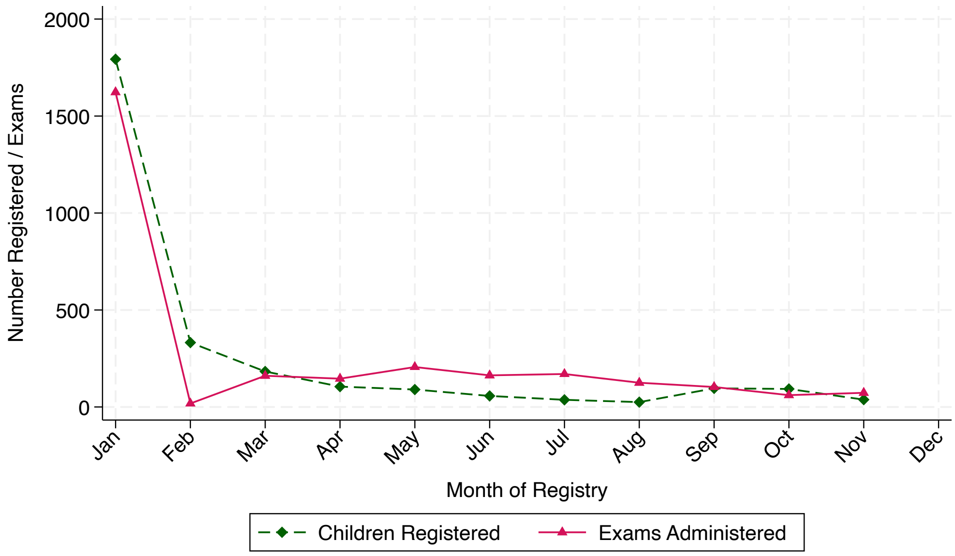
Notes: Standard errors clustered at the country level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Importantly, the coefficients on wait list position and multiple phone numbers remain stable in magnitude and statistical significance after adding shelter residence and gender. This confirms that the capacity and communication barriers are not simply proxying for the socioeconomic vulnerability. Moreover, it implies that capacity and communication improvements would generate absenteeism reductions across the full population of registered families.

Finally, the OLS results in Table 8 are confirmed by the logit estimates reported in Table 10 in the Appendix. The sign, significance, and relative magnitude of all coefficients are consistent across both estimators, and the marginal effects implied by the logit specification are close to the OLS coefficients.

The number of children registered and the number of exams administered per month are both heavily concentrated in January - see Figure 12. This pattern aligns with the timing of conventional school enrollment in Brazil, which follows a calendar that begins in February and runs through December.

Figure 12: Children Registered and Exams Administered per Month

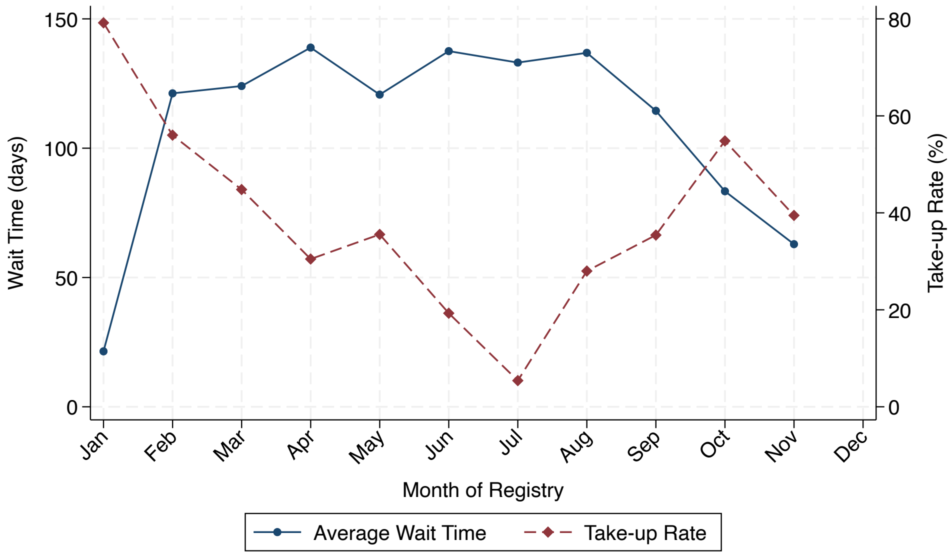


Source: Placement Exam Registry for 2023 and 2024.

However, the number of exams administered in January is not sufficient to meet the high demand in that month. Limited exam administration in February further contributes to a buildup in the waiting list, leading to increasing wait times through August. In the second half of the year, wait times gradually decline, reflecting both the proximity to the next January intake and consistently lower registration levels. This pattern results in an inverted U-shaped trajectory for average wait times, mirroring a U-shaped pattern in registrations (see Figure 13). Therefore, given the Secretariat's capacity in January (1623 exams administered in January 2023 and 2024), matching exams administration with registry in February—and allocating additional attention to registrations outside the conventional school year could reduce waiting times and, consequently, improve take-up.[15]

[15] The increase in registrations from August (25) to September (96) may reflect migrants following the Venezuelan school calendar, which runs from September to June.

Figure 13: Average Wait Time and Take-up per Month of Registry



Source: Placement Exam Registry for 2023 and 2024.

Grade Assignment

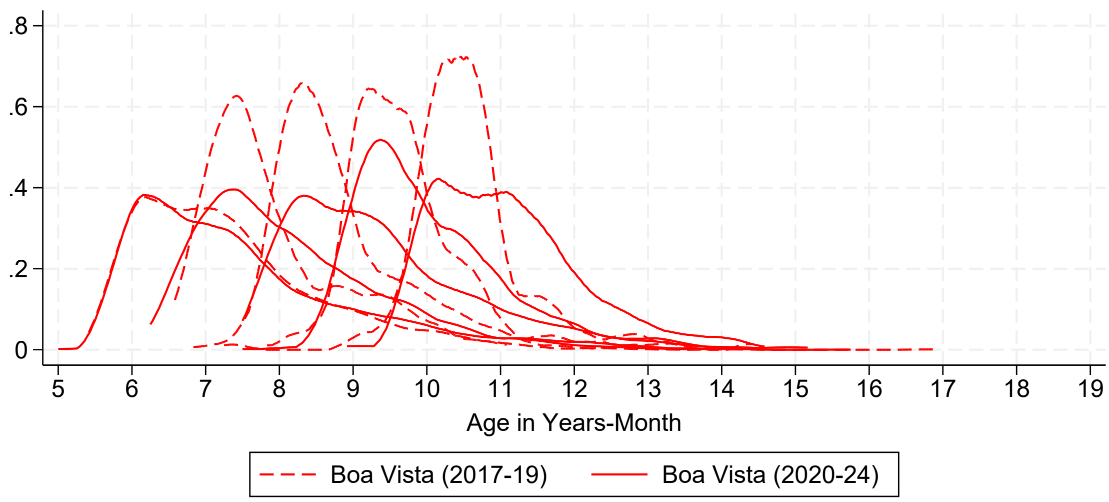
Finally, we examine how grade assignment changed after the policy. More specifically, the policy could also have changed the grades that Venezuelans are assigned to, potentially affecting age distortion rates (share of students at least two years older than expected for that grade) among newly enrolled migrants. The placement exam policy could affect grade assignment/age-distortion through two channels: by altering the selection of children who ultimately enroll, and by imposing a different level of strictness in assessment from the decentralized process it replaced.

Figure 14 plots the age density of newly enrolled Venezuelan students by grade for grades 1 through 5 (Panel a) and grades 6 through 9 (Panel b), comparing the pre (2017–2019) and post-policy (2020–2024) periods. After 2019, the age distribution of children enrolled in each municipal school grade shifts rightward, indicating that children enrolled in grades 2 through 5 are on average older in the post-policy period than in the pre-policy period. This rightward shift is equivalent to an increase in age distortion rates among newly enrolled Venezuelan children in municipal schools. Panel (b), as expected, shows no comparable shift among state schools that were not directly affected by the policy.

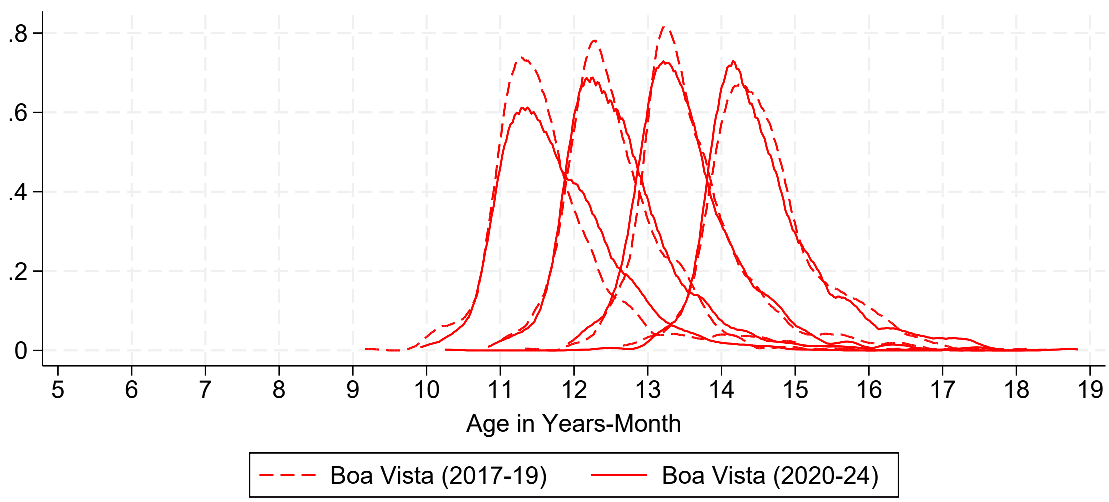
Given that exam absenteeism is higher among sheltered refugees (a particularly vulnerable subgroup of the Venezuelan population) the policy may have selectively excluded lower-performing students. At the same time, because grade assignment prior to the reform was fully decentralized and school-specific, it is unclear whether the previous system was, on average, more lenient or more stringent. As a result, it is not possible to disentangle selection effects from changes in exam strictness. Nonetheless, it is still relevant that the policy increased age-distortion among enrolled students in the early grades (2 to 5) of elementary education.

Figure 14: Age Density Per Grade - Boa Vista (RR)

(a) 1st to 5th Grades



(b) 6th to 9th Grades



Notes: enrollment across all public schools. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

The grade assignment results should be interpreted alongside the negative enrollment results. The children excluded from enrollment by the exam barrier may have been the children who would have experienced the largest age distortion had they enrolled. The observed increase in age distortion among enrolled children may differ from the full grade assignment consequences of the policy. Importantly, age distortion is not necessarily detrimental for refugee students' learning, as placement in lower grades can help address learning gaps, language barriers, and interrupted schooling trajectories. The placement exam effect of grade assignment on enrolled Venezuelan children's learning outcomes in Boa Vista is an open empirical question.

8 Conclusion

Between 2010 and 2024, the number of displaced children worldwide rose from 17 to nearly 50 million. Access to education offers these children stability, protection, and a path to integration, yet schools face challenges such as interrupted schooling, language barriers, and missing documents. Placement exams can assist schools in assessing migrants' pre-enrollment skills and guide grade assignment. However, depending on their design and implementation, they may also function as an unintended barrier, reducing school enrollment.

This paper studies how a mandatory placement exam affected refugee children's school enrollment in Boa Vista, Brazil, following the large inflow of Venezuelan migrants after 2017. The local government introduced a mandatory placement exam at the end of 2019 for all Venezuelan children aged 8–14 without school. Using a triple-difference framework, we exploit both the policy's geographic restriction to Boa Vista and its targeted age group to estimate its impact on school enrollment using national administrative data.

We find that the introduction of the exam in Boa Vista reduced school enrollment among refugee children in the targeted age group. We estimate that the policy affected the enrollment of around 1,000 Venezuelan children between 2020 and 2024. Exploring unique administrative data on the exams' registration list, we also found that the main barrier was not that refugee families failed to register their children for the exam, but rather the waiting time (on average, 30 days) and difficulty in contacting families, which increased the exam absenteeism. Finally, Venezuelan children living in shelters, a specially vulnerable subgroup of the refugee population, were also more likely to miss the exam (by more than 20 percentage points).

This project is among the first economic papers studying the integration of refugee children into the host country's public education. Together, the results reveal that local government should consider changes in the way placement exams are applied. Importantly, improving exam capacity (i.e., reducing waiting times) and communication (such as using alternative strategies, such as publicly available exam schedule lists) can reduce absenteeism and increase enrollment. Conducting placement exams in shelters to reach out-of-school children could also improve enrollment, especially among this vulnerable subgroup of refugees most adversely affected by the policy.

Finally, this paper did not evaluated the effect of grade assignment. More specifically, whether downgrading/age distortion, is beneficial for refugees education outcomes (such as test scores, attendance and retention). Future work in understanding refugee students integration after enrollment is needed.

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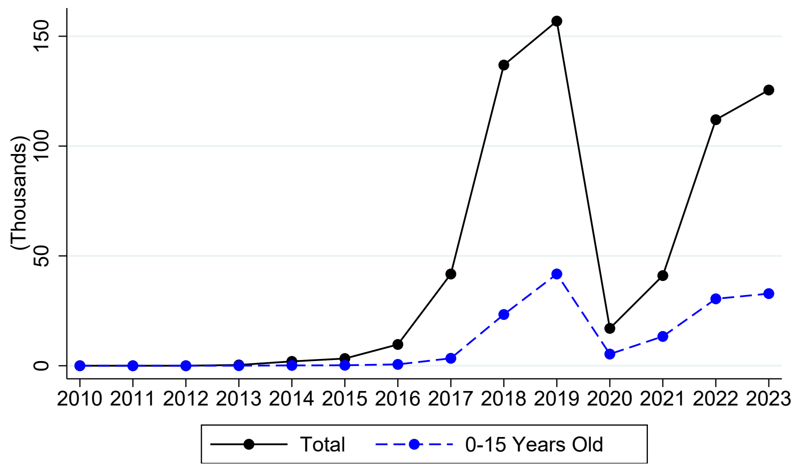
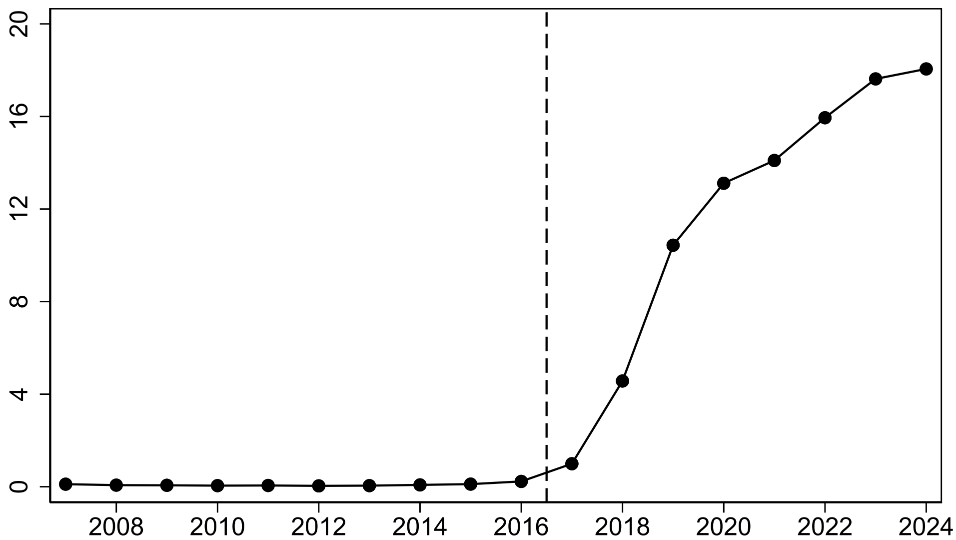
Appendix

Figure 15: Venezuelan Net Entrance Flow - Roraima's Border



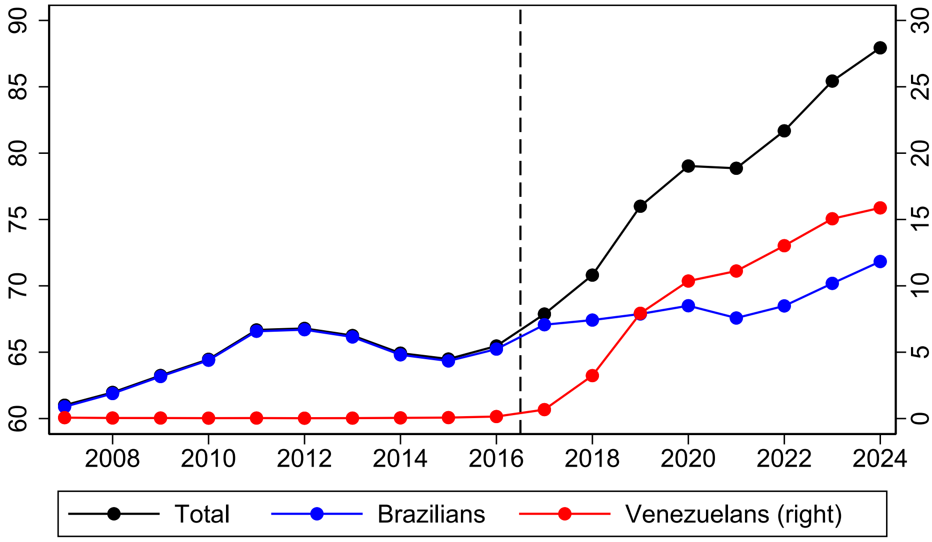
Source: STI.

Figure 16: % Venezuelan Students in Public Schools - Boa Vista (RR)



Source: School Census (2017-2024).

Figure 17: Number (1,000) of Venezuelan Students in Public Schools - Boa Vista (RR)



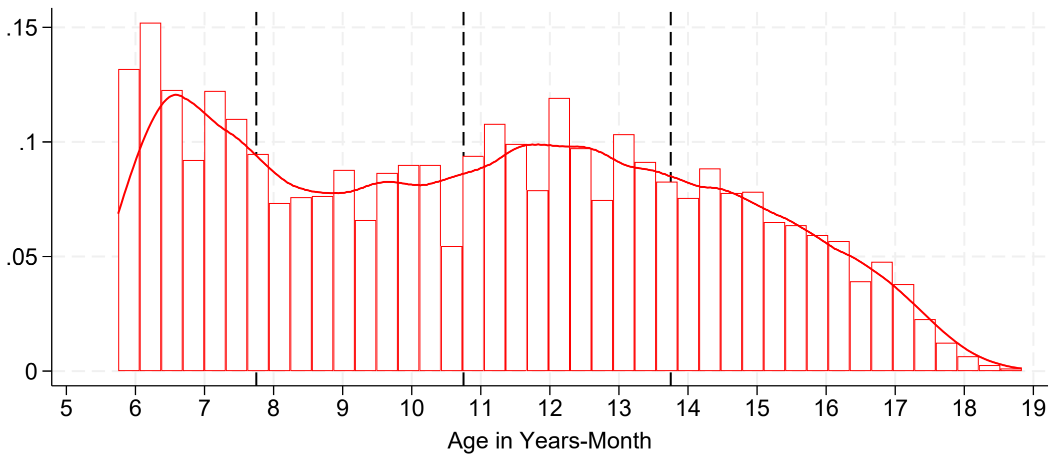
Note: Includes all public schools from grades 1 to 12, kindergarten to Nursery schools. *Source:* School Census (2017-2024).

Table 9: Exam Registry, Take-up, and Enrollment (2023 and 2024)

Age	**In the Registry**	**Took the Exam**	**Enrolled**
8	481	350	319
9	542	377	344
10	669	507	468
11	557	415	393
12	248	165	156
13	109	69	53
14	57	30	7
Total	**2,663**	**1,913**	**1,740**

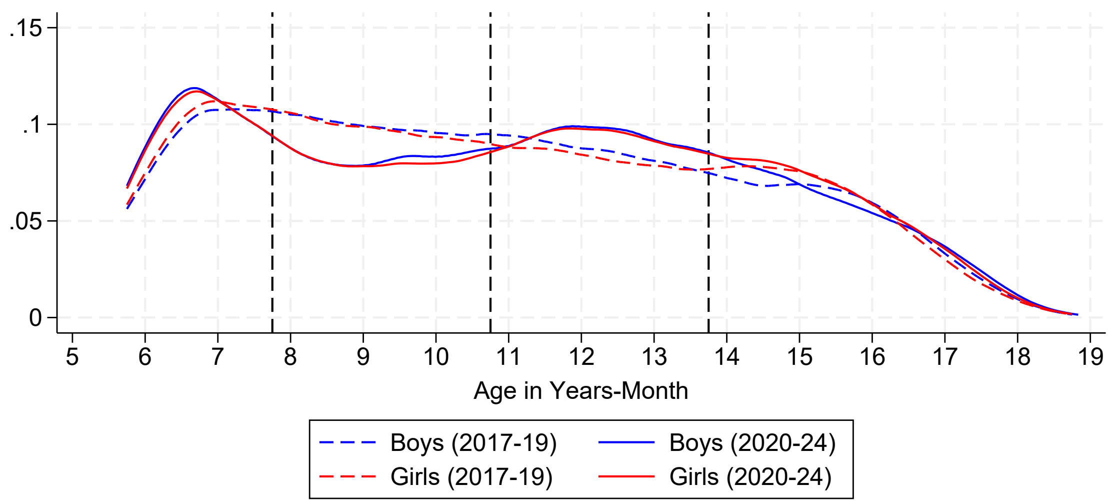
Notes: Enrollment is inferred by matching exam takers to newly enrolled Venezuelan students with the same birth date in the expected year of enrollment based on the date of the exam in the administrative data. *Source:* administrative placement exam registry (Boa Vista´s Secretary of Education) and the administrative national enrollment panel (INEP).

Figure 18: Age Histogram of Newly Enrolled Venezuelans - Boa Vista (RR)



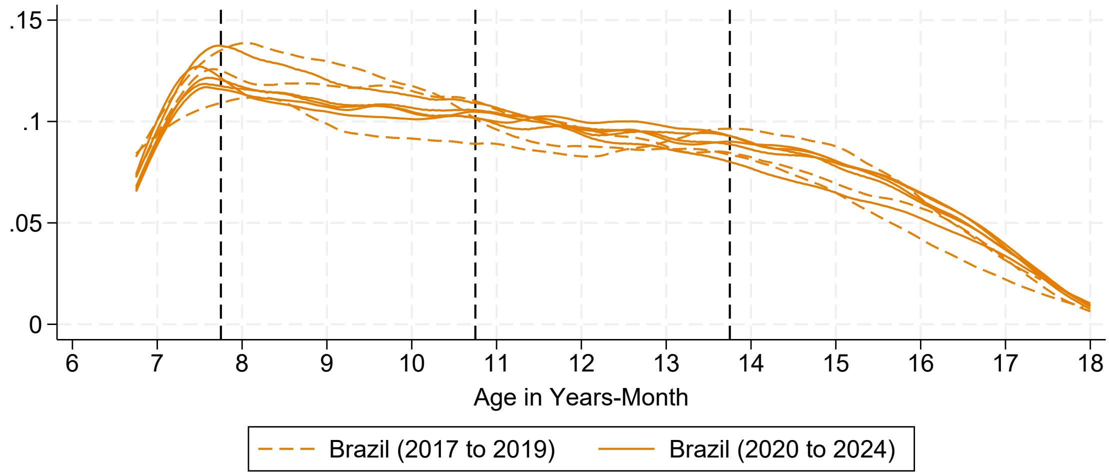
Notes: enrollment across all public schools. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

Figure 19: Age PDF of Newly Enrolled Venezuelans (Gender) - Boa Vista (RR)



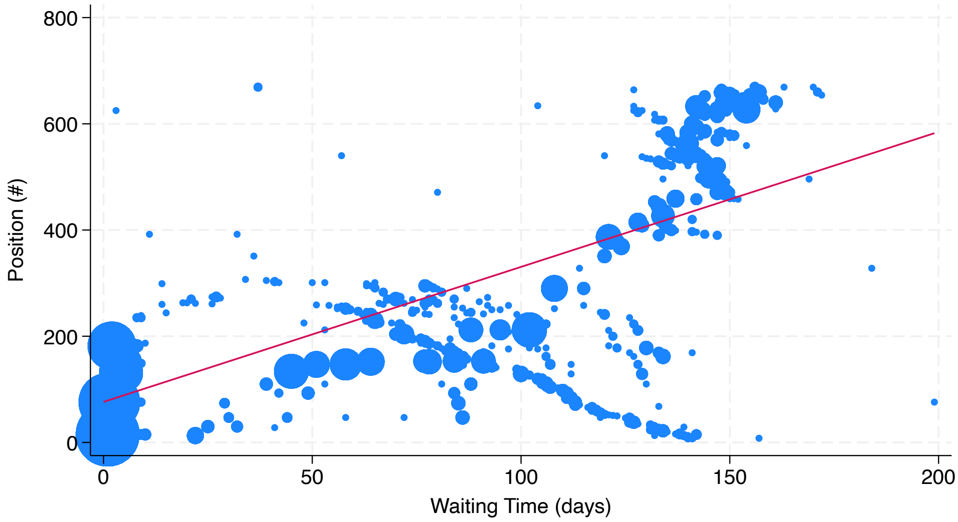
Notes: enrollment across all public schools. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

Figure 20: Age PDF of Newly Enrolled Venezuelans (Yearly) - Brazil (excluding RR)



Notes: enrollment across all public schools in Brazil excluding Roraima. PDF estimated using an univariate kernel density estimation. *Source:* School Census (2017-2024).

Figure 21: Position in the Wait List Vs Waiting Time



Note: red curve represents the fit/regression line between the two variables and a constant. *Source*: Placement Exam Registry for 2023 and 2024.

Table 10: What Explains Exam Absenteeism - Logit

Eq. 5	Y = *Missed Exam*$_i$				
	(1)	(2)	(3)	(4)	(5)
Wait List Position	0.0031***	0.0031***	0.0038***	0.0044***	0.0045***
	(0.0003)	(0.0003)	(0.0007)	(0.0009)	(0.0011)
Have Phone Number	-0.8201**	-0.8453**	-0.8061**	-1.3074	-1.1849
	(0.3440)	(0.3505)	(0.3432)	(0.9028)	(0.9670)
Have 2+ Phone Numbers	-0.5880***	-0.5723***	-0.5194***	-0.4213**	-0.5530**
	(0.1393)	(0.1409)	(0.1535)	(0.1841)	(0.2211)
Have NGO Phone Number	0.9775***	1.0179***	1.1774***	0.2808	0.3402
	(0.1630)	(0.1619)	(0.1762)	(0.2364)	(0.2874)
Live in a Shelter				1.1905***	1.3434***
				(0.1852)	(0.2168)
Girl					0.0533
					(0.1572)
Observations	3,298	3,298	2,785	1,375	1,013
Fixed Effects:					
Year FE	Y	Y	N	N	N
Age FE	N	Y	Y	Y	Y
Month-Year Registered FE	N	N	Y	Y	Y
Avg Missed Exam	0.299	0.299	0.354	0.531	0.536
Avg. Position Wait List	222.2	222.2	220.1	331.5	336.9
Avg. Have a phone	0.980	0.980	0.978	0.993	0.992
Avg. Have 2+ phones	0.130	0.130	0.118	0.142	0.139
Avg. Have NGO phone	0.0797	0.0797	0.0822	0.145	0.141
Avg. Live in Shelter				0.324	0.324

Notes: Standard errors clustered at the country level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.